

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I Bala S (192524427) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: bala

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

1. Explain the difference between HDFS and traditional file systems?

HDFS	Traditional File System
Distributed across many nodes	Usually local/centralized
Designed for very large files	Designed for general-purpose files
Provides automatic replication	Replication is usually not built-in
Fault tolerant	Depends on hardware/system backup
High throughput	Optimized more for general file access

2. Why is replication important in distributed storage systems?

- Replication means storing multiple copies of the same data on different nodes.
- For example, with replication factor = 3, three copies of a data block are stored on different nodes.

Benefits:

- Prevents data loss when a node fails.
- Maintains data availability.
- Allows processing to continue during hardware failures.
- Improves read performance because data can be read from multiple locations.

3. How does MapReduce divide work between map and reduce phases?

MapReduce divides large data-processing jobs into two main phases:

Map phase:

- Input data is divided into blocks/splits.
- Multiple mapper tasks process these splits in parallel.
- Each mapper produces **key-value pairs**.

Shuffle and Sort:

- Intermediate key-value pairs are grouped by key.
- Data belonging to the same key is sent to the same reducer.

Reduce phase:

- Reducers process grouped data.
- They generate the final output.

4. What role does YARN play in large-scale Hadoop processing?

YARN (Yet Another Resource Negotiator) is Hadoop's resource-management and job-scheduling layer.

Its main responsibilities are:

- Allocating CPU and memory to applications.
- Scheduling jobs across cluster nodes.
- Monitoring running applications.
- Managing multiple workloads simultaneously.

5. Differentiate NAS and SAN in an enterprise data environment

NAS	SAN
Network Attached Storage	Storage Area Network
Provides file-level storage	Provides block-level storage
Accessed using protocols such as NFS/SMB	Commonly uses Fibre Channel/iSCSI
Easier to deploy and manage	More complex and expensive
Suitable for file sharing and backups	Suitable for databases and high-performance applications

6. What is the purpose of ETL in a big data pipeline?

ETL = Extract, Transform, Load.

- Extract: Collect data from databases, applications, APIs, logs, etc.
- Transform: Clean, validate, filter, format, and integrate the data.
- Load: Store the processed data in a target system such as HDFS, a data warehouse, or a data lake.

Example:

Customer Database + Web Logs → Extract → Clean/Transform → HDFS/Data Warehouse

7. How does Apache NiFi help in data integration workflows?

Apache NiFi is a data-flow and integration tool that automates the movement of data between different systems.

- Ingest data from databases, APIs, files, and IoT devices.
- Transform and route data.
- Connect different systems.
- Monitor data flows.
- Provide data provenance and tracking.
- Handle failures and back-pressure.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

1. Data Node failure

- HDFS detects missing replicas.
- It automatically re-replicates blocks.

2. Name Node failure

- Use Name Node High Availability (HA) with Active and Standby Name Nodes.

3. Resource Manager failure

- Use YARN Resource Manager HA.

4. Network failure

- Use redundant network paths and proper rack-aware configuration.

5. Disk failure

- Monitor disk health and maintain multiple replicas.

6. Application/job failure

- Use task retries, monitoring, and logging.

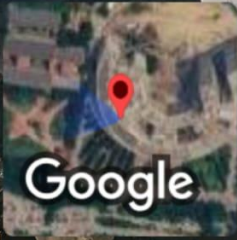
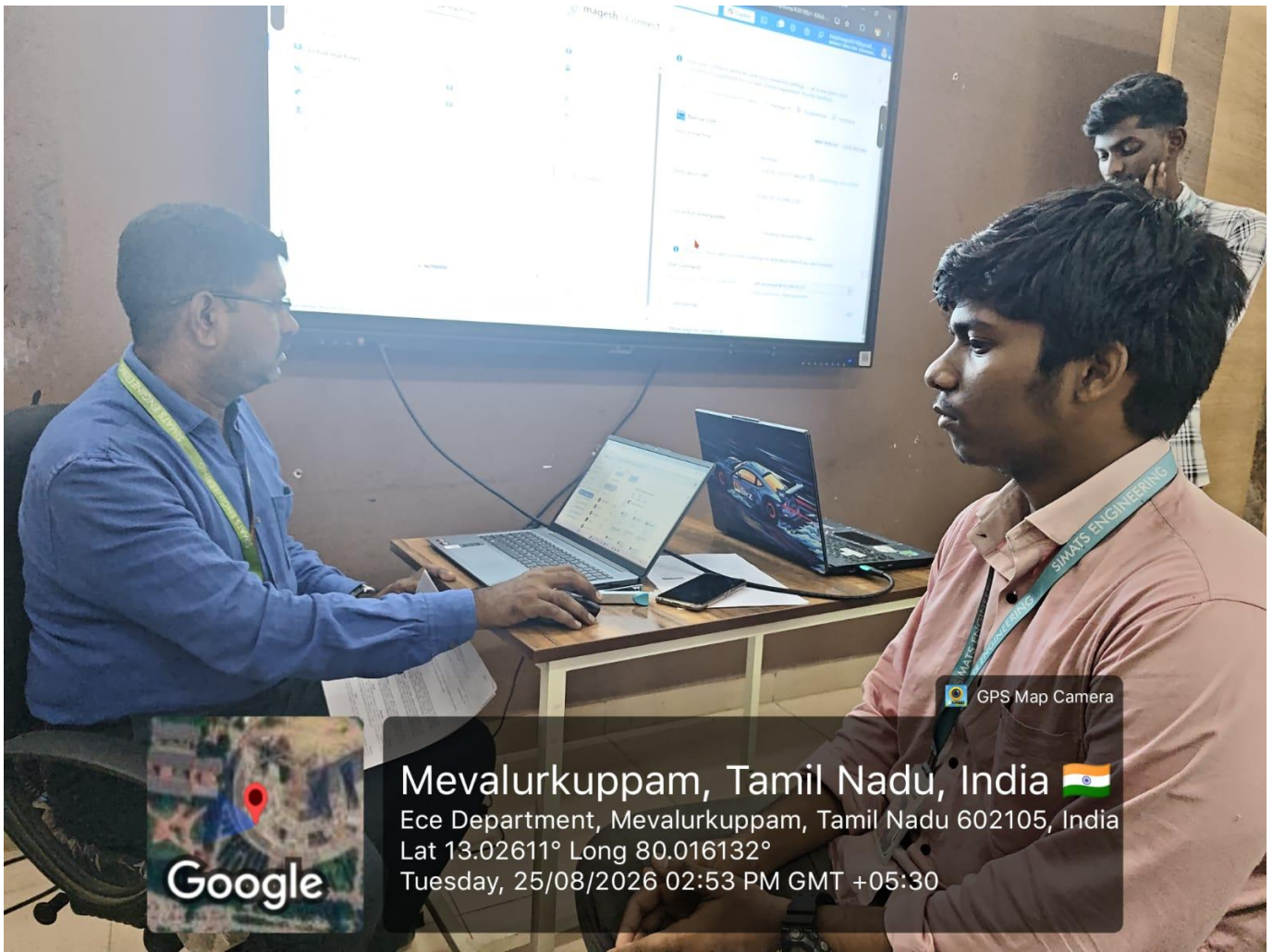
9. How would you design a secure data ingestion pipeline for a large organization?

- Use TLS/SSL for data transmission.
- Authenticate users and services using Kerberos where appropriate.
- Apply role-based access control.
- Encrypt sensitive data at rest.
- Validate and sanitize incoming data.
- Maintain audit logs and data provenance.
- Use network segmentation and firewalls.
- Apply least-privilege access.
- Monitor the pipeline for suspicious activity.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Business benefits:

- Handles very large data volumes.
- Reduces dependence on one machine.
- Continues operating despite hardware failures.
- Allows horizontal scaling by adding servers.
- Provides high-throughput data processing.



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